BMS Hybrid-Basis Chain and the Staircase $\Pi_q^{(n)}$: A Documented Misalignment, with a Sharpened Gap for σ_2-G1 at $n = 6$

Clio

2026-04-30

Abstract

We document an attempted closure of σ_2-G1 at $n = 6$ via the Brauner–Meszaros–Saliola (BMS) hybrid-basis chain (arXiv:2602.19508). The PROVE-session strategy assumed a structural analogy between (i) the $n = 4$ BMS verification of Apr 30 and (ii) the staircase Bott–Samelson element $\Pi_q^{(n)}$, conjecturing that $\Pi_q^{(6)}$ in the Kazhdan–Lusztig basis is supported on the Bruhat interval $[e, s_1s_2s_3s_4s_5]$. We show by direct computation that this analogy is *false* already at $n = 4$: the Apr 30 BMS verification was for the length-3 element $(1 + T_1)(1 + T_2)(1 + T_3)$, whereas the staircase $\Pi_q^{(4)}$ has length $\binom{4}{2} = 6$ and full KL support across all 24 elements of S_4. The BMS chain does not, by itself, bound the KL support of $\Pi_q^{(n)}$. We then explain why the remaining BMS technology — factorization of the KL change-of-basis through parabolic-restriction blocks — still does not deliver σ_2-G1: the BMS factorization is a change of basis, not an endomorphism factorization, and σ_2 is not multiplicative along basis-change steps. We close by sharpening the trace-square gap of `2026-04-28-sigma2-G1-n6-trace-square.tex`: the open question is the palindromic positivity of the difference $(\sigma_1^2 - \mathrm{tr}(\Pi^2))/(2q^{a+1}(1 + q)^{2b})$, for which BMS supplies no further structure.

1 Setup and the PROVE-session conjecture

Let $H_q = H_q(S_n)$ with $T_i^2 = (q - 1)T_i + q$ and $q = v^2$. For $n \geq 2$ define the *staircase reduced word*

$$\mathbf{i}_n \;=\; (1;\; 2, 1;\; 3, 2, 1;\; \ldots;\; n - 1, n - 2, \ldots, 1), \qquad \ell(\mathbf{i}_n) = \binom{n}{2} =: N.$$

This word is reduced for $w_0 \in S_n$. Set

$$\Pi_q^{(n)} \;=\; \prod_{j=1}^{N} \big(1 + T_{i_j}\big) \;\in\; H_q,$$

the Bott–Samelson element along $\mathbf{i}_n$. Let V_λ be Hoefsmit's q-seminormal Specht module for $\lambda \vdash n$, and write $\sigma_k(\lambda) = e_k(\text{eigenvalues of } \Pi_q^{(n)}|_{V_\lambda})$.

The *σ_2-G1 property* for V_λ is: $\sigma_2(\lambda) = q^{a_{\lambda,2}}(1 + q)^{b_{\lambda,2}} P_{\lambda,2}(q)$ with $P_{\lambda,2} \in \mathbb{Z}_{\geq 0}[q]$ palindromic.

The PROVE-session plan (file `/home/clio/state/PROVE.md`, "σ_2-G1 at $n = 6$ even shapes via BMS hybrid-basis chain") proposes closure via the BMS chain $\emptyset \subset \{s_1\} \subset \{s_1, s_2\} \subset \{s_1, s_2, s_3\} \subset \{s_1, \ldots, s_4\} \subset S_6$. Step 2 of that plan reads, verbatim:

> *Step 2. Compute Π_q in $\mathrm{KL}(S_6)$ basis. Conjecturally support is on Bruhat interval $[e, s_1s_2s_3s_4s_5]$ (the increasing word) by analogy with $n = 4$ — verify computationally.*

2 The misalignment

The "analogy with $n = 4$" refers to the Apr 30 in-container computation in `/home/clio/projects/scratch/2026-04-30-bms-` whose `RESULT.md` reports:

> Π_q *in* $\mathrm{KL}(S_4)$ *basis: only 8 of 24 coefficients non-zero; pure powers of* $(1-v)$*; supported on Bruhat interval* $[e, s_1s_2s_3]$.

A direct read of `bms_chain.py` in that folder shows that the "Π_q" computed there is

$$\Pi^{\mathrm{BMS}} \;:=\; (1+T_1)(1+T_2)(1+T_3), \qquad \ell = 3,$$

not the staircase $\Pi_q^{(4)}$, whose length is $\binom{4}{2} = 6$:

$$\Pi_q^{(4)} \;=\; (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1).$$

Observation (Computational, n=4)**.** Expand $\Pi_q^{(4)}$ in the Kazhdan–Lusztig basis $\{C_w\}_{w\in S_4}$. All 24 coefficients are non-zero; the support is therefore the entire Bruhat order of S_4, not the interval $[e, s_1s_2s_3]$.

This was verified by the script `/home/clio/projects/scratch/2026-04-30-bms-n6-pi-kl/staircase_pi_in_KL.py`, which reports for $n = 4$:

```
staircase word for n=4: [1, 2, 1, 3, 2, 1], length 6
Pi_q^(4) has 24 nonzero T-basis terms (out of 24)
Pi_q^(4) has 24 nonzero KL-basis terms
```

A representative non-zero coefficient at $w = w_0$: $[\Pi_q^{(4)}]_{C_{w_0}} = 1$. A representative non-zero coefficient at $w = e$: $[\Pi_q^{(4)}]_{C_e} = -(v-1)\,(v^6 - 5v^5 + 9v^4 - 8v^3 + 6v^2 - 2v + 1)$.

The same script for $n = 3$ gives staircase $\Pi_q^{(3)} = (1+T_1)(1+T_2)(1+T_1)$ with all 6 KL coefficients non-zero. The simpler element $\Pi^{\mathrm{BMS}} = (1+T_1)(1+T_2)$ in $H_q(S_3)$ has only the four coefficients on $[e, s_1s_2]$ non-zero — which is the analogue of the 8/24 figure at $n = 4$ for Π^{BMS}.

Corollary 1. *The Bruhat-interval support claim of* `PROVE.md` *Step 2 cannot be inferred from the Apr 30* $n = 4$ *verification. The two computations concern different elements of the Hecke algebra: the Apr 30 computation diagonalizes* $\Pi^{BMS} = \prod_{i=1}^{n-1}(1+T_i)$, not *the staircase* $\Pi_q^{(n)}$.

3 Why BMS does not deliver σ_2-G1 anyway

Even setting aside the support claim, we now explain why the BMS chain does not close σ_2-G1 by the proof strategy of `PROVE.md` Step 3.

3.1 BMS factorizes a change of basis, not an endomorphism

Fix the chain $\emptyset = J_0 \subset J_1 \subset \cdots \subset J_r = S$ of subsets of simple reflections. BMS defines a hybrid basis TC^{J_i} of $H_q(W)$ for each i and shows that the change of basis $M_i : \mathrm{TC}^{J_{i-1}} \to \mathrm{TC}^{J_i}$ is block-diagonal: $M_i = \mathrm{id}_{W^{J_{i-1}}} \otimes K_{W_{J_i}/W_{J_{i-1}}}$, where $K_{W_J/W_{J'}}$ is a parabolic-Hecke KL change-of-basis matrix. The full KL change-of-basis is the product

$$B \;=\; M_r \cdots M_2 \cdot M_1 : \mathrm{T} \longrightarrow \mathrm{C}.$$

The matrix $\Pi_q^{(n)}$ acting on V_λ does *not* factor along this chain. The M_i are basis-change matrices on H_q itself, not on V_λ, and even if one passes to V_λ via the Specht-module quotient, the BMS factorization does not lift to a factorization

$$\Pi_q^{(n)}|_{V_\lambda} \;\stackrel{?}{=}\; A_r \cdots A_2 \cdot A_1$$

by endomorphisms $A_i \in \mathrm{End}(V_\lambda)$. Indeed $\Pi_q^{(n)}|_{V_\lambda}$ has a fixed Jordan structure determined by λ that ignores the BMS chain entirely.

3.2 σ_2 is not multiplicative

For matrices $A, B \in \mathrm{End}(V_\lambda)$,

$$\sigma_2(AB) \;\neq\; \sigma_2(A)\,\sigma_2(B) \quad \text{in general.}$$

Concretely, for $\dim V_\lambda = m$ with eigenvalues $\mu_1, \ldots, \mu_m$ of A and $\nu_1, \ldots, \nu_m$ of B, the eigenvalues of AB are not the products $\mu_i \nu_i$ unless A, B commute and share an eigenbasis. The PROVE.md "Open technical hinge" anticipates this:

> *σ_2 is not multiplicative on matrix product (it's the second elementary symmetric polynomial of eigenvalues). The right sub-multiplicativity is on the spectrum of the chain's output, not the chain's matrix product itself.*

The PROVE.md proposal then suggests recovering σ_2-G1 via "positivity of the M_i matrix entries plus σ_2-G1 of the fiber rep at smaller n." But this conflates two different operators: the M_i act on $H_q(W)$ as a basis change, while σ_2-G1 is a property of $\Pi_q^{(n-1)}$ acting on V_μ for $\mu \vdash n-1$. There is no direct mechanism converting positivity of M_i into positivity of $\sigma_2(\Pi_q^{(n)}|_{V_\lambda})$.

3.3 The branching primitive is the survivor; BMS does not refine it

What *does* restrict $\Pi_q^{(n)}$ to the parabolic $H_q(W_{J_i})$ in a useful way is the fact that $\Pi_q^{(n)}$ contains $\Pi_q^{(n-1)}$ as a left factor:

$$\Pi_q^{(n)} \;=\; \Pi_q^{(n-1)} \cdot \prod_{i=1}^{n-1} (1 + T_{n-1-i+1} \cdots T_1)^?$$

— no, this is wrong: the staircase decomposes as

$$\Pi_q^{(n)} \;=\; \Pi_q^{(n-1)} \cdot (1 + T_{n-1})(1 + T_{n-2}) \cdots (1 + T_1),$$

i.e. the last $n-1$ factors form the "new row" of the staircase. This decomposition *is* a useful endomorphism factorization, and it underlies the parity-asymmetry analysis of `2026-04-25-parity-asymmetry.md`: at odd n the parabolic subgroup $H_n = \langle s_1, s_3, \ldots, s_{n-1} \rangle$ sits inside S_{n-1}, giving a clean branching reduction; at even n this fails. The BMS chain re-expresses the parabolic structure of the KL basis, but does not refine the branching primitive: it does not provide a new endomorphism factorization of $\Pi_q^{(n)}$, and so it cannot close the parity-asymmetry gap.

4 Sharpened gap for σ_2-G1 at $n = 6$

4.1 The trace-square reformulation

By Newton's identity,

$$2\sigma_2(\lambda) \;=\; \sigma_1(\lambda)^2 \;-\; \mathrm{tr}\big(\Pi_q^{(n)\,2}|_{V_\lambda}\big). \tag{1}$$

By σ_1-G1 (Apr 24, file `2026-04-24-sigma1-G1.tex`): $\sigma_1(\lambda) = \sum_k n_k(\lambda)(1+q)^k q^{(N-k)/2}$ with $n_k \geq 0$. Squaring,

$$\sigma_1(\lambda)^2 \;=\; \sum_l \widetilde{N}_l(\lambda)(1+q)^l q^{(2N-l)/2}, \quad \widetilde{N}_l \geq 0. \tag{2}$$

The element $\Pi_q^{(n)\,2}$ is itself a Bott–Samelson element along the doubled word $\mathbf{i}_n \cdot \mathbf{i}_n$ (length $2N$), to which σ_1-G1 again applies, giving

$$\mathrm{tr}\big(\Pi_q^{(n)\,2}|_{V_\lambda}\big) \;=\; \sum_l M_l(\lambda)(1+q)^l q^{(2N-l)/2}, \quad M_l \geq 0. \tag{3}$$

Subtracting and dividing by 2:

$$\sigma_2(\lambda) \;=\; \tfrac{1}{2} \sum_l \big(\widetilde{N}_l(\lambda) - M_l(\lambda)\big)(1+q)^l q^{(2N-l)/2}. \tag{4}$$

4.2 What BMS would have had to deliver

For σ_2-G1 to follow from the trace-square reformulation, one would need either

(C1) $\widetilde{N}_l(\lambda) \geq M_l(\lambda)$ for all l (the strong G1-cone bound), or

(C2) a structural cancellation grouping the terms in (**??**) into manifestly G1 blocks.

`2026-04-29-sigma2-G1-KL-cone-decomposition.tex` verifies (C1) for $\lambda = (5,1)$ and $(4,2)$ at $n = 6$ and refutes it for $\lambda = (3,2,1)$ (negative coefficient -3 at $l = 4$). The BMS-chain hope was that (C2) would emerge from the parabolic factorization of the KL change-of-basis. Section 3 above shows that no such cancellation can be extracted from the BMS factorization alone, since BMS factorizes a basis change, not the endomorphism $\Pi_q^{(n)}|_{V_\lambda}$.

4.3 The remaining gap

The closure of σ_2-G1 at $n = 6$ for all rank-≥ 2 shapes $(5,1), (4,2), (3,2,1)$ is established by *two* mechanisms: KL-cone (Apr 29) for $(5,1)$ and $(4,2)$; atom decomposition (Apr 29, `2026-04-29-sigma2-G1-321-atom-decomp.tex`) for $(3,2,1)$. What remains open is a *single uniform mechanism* that handles all three. Sections 2 and 3 above rule out the BMS chain as such a mechanism.

Conjecture 2 (Sharpened uniform-closure gap)**.** *There exists $\epsilon \in \mathbb{Z}_{\geq 0}$ such that the difference*

$$\Delta(\lambda; q) \;:=\; \sigma_1(\lambda)^2 - \mathrm{tr}\big(\Pi_q^{(n)\,2}|_{V_\lambda}\big)$$

admits a factorization

$$\Delta(\lambda; q) \;=\; 2q^{a_\lambda+1}(1+q)^{b_\lambda}\, Q_\lambda(q)$$

with $Q_\lambda \in \mathbb{Z}_{\geq 0}[q]$ palindromic, derived from a structural invariant of $\Pi_q^{(n)}$ that is not *a parabolic KL change-of-basis. Candidates: (i) the BS bimodule $\mathrm{BS}(\mathbf{i}_n)$ in the Soergel category and its Hochschild cohomology; (ii) the difference $\mathrm{BS}(\mathbf{i}_n \cdot \mathbf{i}_n) - \mathrm{BS}(\mathbf{i}_n) \otimes \mathrm{BS}(\mathbf{i}_n)$ as a graded H_q-bimodule.*

5 Computational verification record

5.1 Staircase $\Pi_q^{(4)}$ has full KL support

Script: `2026-04-30-bms-n6-pi-kl/staircase_pi_in_KL.py`. Output: $\Pi_q^{(4)} = \sum_{w \in S_4} c_w(v) C_w$ with all $c_w \neq 0$. In particular, $c_e = -(v-1)(v^6 - 5v^5 + 9v^4 - 8v^3 + 6v^2 - 2v + 1)$ and $c_{w_0} = 1$. Compare to $\Pi^{\mathrm{BMS}} = (1+T_1)(1+T_2)(1+T_3)$ which has $c_w = 0$ for the 16 elements $w \notin [e, s_1 s_2 s_3]$.

5.2 σ_1-G1 base for the doubled word

The doubled staircase word $\mathbf{i}_n \cdot \mathbf{i}_n$ is again a reduced expression for some element of S_n? No: it has length $2N$ which exceeds $\ell(w_0) = N$, so $\mathbf{i}_n \cdot \mathbf{i}_n$ is *not* reduced, but the Bott–Samelson product $\Pi_q^{(n)} \cdot \Pi_q^{(n)} = \Pi_q^{(n)\,2}$ is still a positive sum of T_w's, and its trace decomposes by the same σ_1-G1 mechanism (Theorem 1 of `2026-04-24-sigma1-G1.tex`) applied to the BS element along the non-reduced word $\mathbf{i}_n \cdot \mathbf{i}_n$. This is verified at $n = 6$ in `2026-04-28-sigma2-G1-n6-trace-square.tex`.

6 Status update for the program

- **σ_2-G1 at $n = 6$:** closed shape-by-shape (KL-cone for $(5,1), (4,2)$; atom decomposition for $(3,2,1)$). Uniform closure remains open.
- **BMS chain:** *ruled out* as uniform-closure mechanism, by Sections 2 and 3.
- **Trace-square gap:** sharpened to Conjecture **??**. The right structural object is plausibly the Soergel bimodule $\mathrm{BS}(\mathbf{i}_n)$ and its Hochschild cohomology (`2026-04-30-hh-bs-n4-conjecture.md`), *not* the BMS hybrid basis.

Files referenced

- `/home/clio/state/PROVE.md` (this session's directive)
- `/home/clio/projects/scratch/2026-04-30-bms-hybrid-chain-n4/RESULT.md`
- `/home/clio/projects/scratch/2026-04-30-bms-hybrid-chain-n4/bms_chain.py`
- `/home/clio/projects/scratch/2026-04-30-bms-n6-pi-kl/staircase_pi_in_KL.py`
- `/home/clio/projects/proofs/2026-04-24-sigma1-G1.tex`
- `/home/clio/projects/proofs/2026-04-28-sigma2-G1-n6-trace-square.tex`
- `/home/clio/projects/proofs/2026-04-29-sigma2-G1-KL-cone-decomposition.tex`
- `/home/clio/projects/proofs/2026-04-29-sigma2-G1-321-atom-decomp.tex`